



Technical Service Bulletin

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Alternator Issues in Arctic Conditions

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Core Issue

Delco-Remy™ 33SI alternators (Part Number 3400698) are experiencing rear bearing failures in arctic conditions, where the ambient air temperature is less than -5°C [23°F].

Confirmation

All K19 and QSK19 engines built with Delco-Remy™ 33SI alternators.

Customers are reporting that there is no charge to the battery, and on investigation, have found the rear bearing broken with progressive damage to the stator and rotor.

The alternator will **not** produce any charge when the engine is operating.

The grease and rear bearing used in the manufacture of the 33SI alternator is more susceptible to arctic weather conditions.

The rear bearing failure damages the alternator, resulting in replacement of the alternator with a new unit or an attempted field repair.

Resolution

Replace failed Delco-Remy™ 33SI alternators with the Prestolite BLD 3314GH alternator, as the Prestolite alternator is more capable of coping with extreme arctic temperatures of -40°C [-40°F].

All 19L engines built after November 2008 for arctic applications now include the Prestolite BLD 3314GH alternator and the required change of pulleys, belt, and ground wire.

Alternator Options			
Delco-Remy™ 33SI		Prestolite BLD 3314GH	
Description	Part Number	Description	Part Number
Delco-Remy™ 33SI Alternator	3400698	Prestolite BLD 3314GH Alternator	3975140
5-Groove Alternator Pulley	3002323	9-Groove Alternator Pulley	4953300
5-Groove Alternator Drive Pulley	3002331	9-Groove Alternator Drive Pulley	4009258
Cummins® 5-rib Belt	206996	Cummins® 9-rib Belt	3024614

Alternator Options			
Delco-Remy™ 33SI		Prestolite BLD 3314GH	
Ground Wire	3202162	N/A	N/A

When upgrading to the Prestolite alternator, the ground wire will have to be changed as the Prestolite BLD 3314GH has a higher output than the Delco-Remy™ 33SI. The ground wire is presently **not** available for purchase from Cummins Inc.

There are no changes to the mounting bracket, tensioning arm, or the installation/removal and belt tension procedures, and no service tools required.

The specification for the ground wire is:

2 AWG

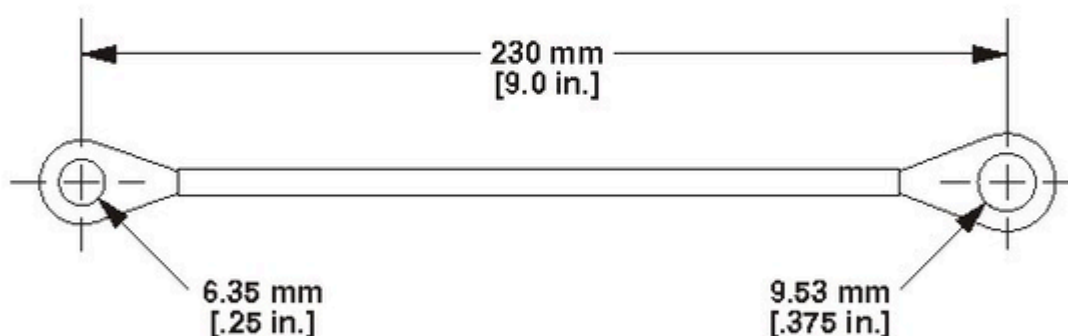
230 mm [9 in] long

One eyelet with an internal diameter of 6.35 mm [1/4 in]

One eyelet with an internal diameter of 9.53 mm [3/8 in]

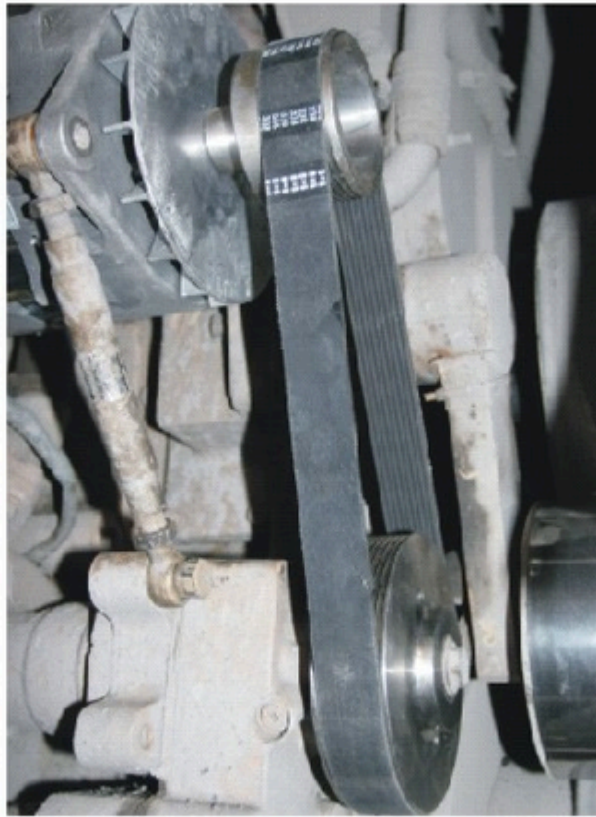
This is shown in the photograph below.

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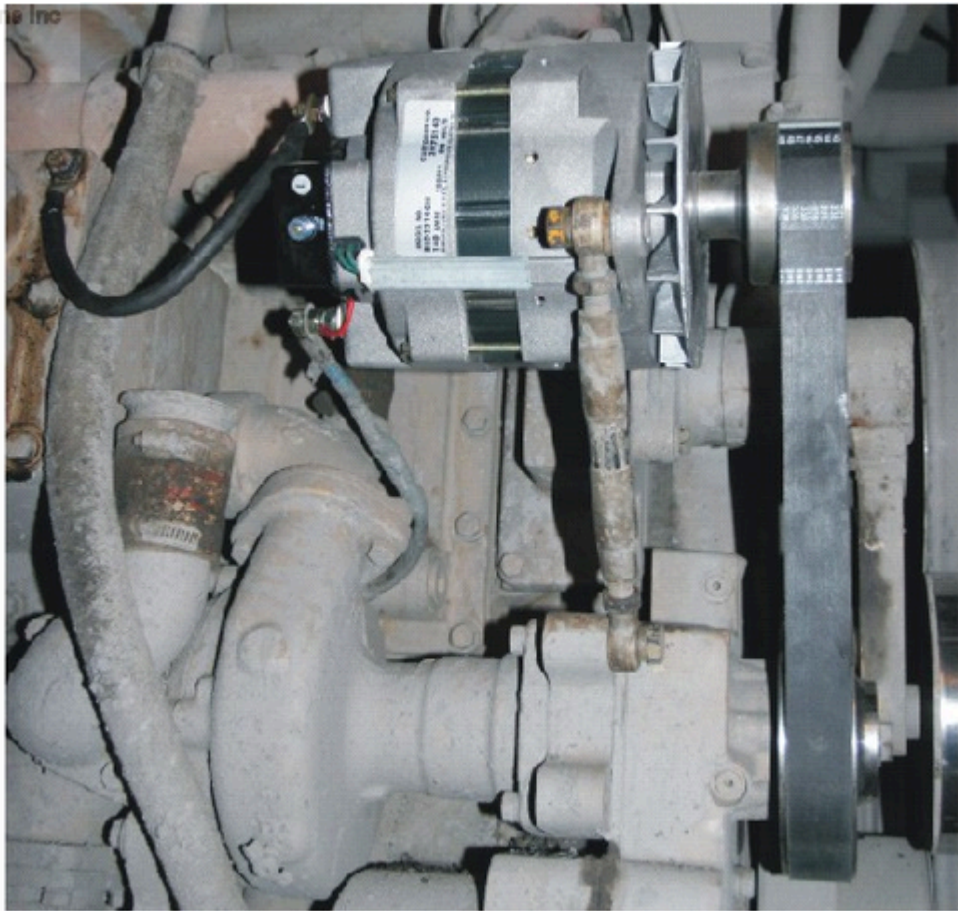
Photographs of an installed Prestolite BLD 3314GH alternator.



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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Last Modified: 08-Dec-2008
